Search and Optimization in BISECT: A U-Series Overview

Gio Della-Libera

Draft — May 13, 2026

Abstract

This draft organizes the BISECT U-series after the RPLAN fixed point. It separates search, exact optimization, multi-objective selection, legal posture, and audit/certification roles so that later papers can make claims at the right evidence level. RPLAN/RCTX artifacts are treated as reproducibility and lineage boundaries, not as proofs of political fairness or legal compliance.

1 Introduction

The U-series covers the methods that choose, improve, certify, and compare candidate plans after construction. This overview is a taxonomy paper: it does not add new algorithms or empirical results, but organizes the existing U.1-U.20 surface so implementation, evidence, and legal interpretation stay separated.

The practical contract is that every method eventually emits or consumes a declared plan artifact with enough context for downstream audit. In the current implementation program, that boundary is the RPLAN/RCTX fixed point: a final plan, the run context that produced or selected it, and the audit sidecars that state which profile was checked. U.0 therefore reads the U papers as a family of roles around a common artifact rather than as competing claims of equal kind.

2 Taxonomy

The track is organized around five roles: stochastic search and ensembles, exact optimization, multi-objective selection, legal posture, and audit certification. The shared RPLAN/RCTX fixed point is the artifact boundary that lets heterogeneous methods be compared without pretending they prove the same kind of claim.

The taxonomy is deliberately role based. A branch-and-cut run, a large neighborhood search run, and a Pareto-front selection run may all emit usable plans, but their evidence differs: one may offer a bound or infeasibility witness, another may offer a validity-preserving improvement trace, and another may offer a documented trade-off choice among non-dominated alternatives.

Paper band	Role	Primary claim type
U.1–U.11	Existing search and ensemble foundations	Method and empirical framing
U.12	Algorithm-selection matrix	Methodology and operator guidance
U.13	Exact-vs-heuristic certification	Certificate interpretation
U.14	Multi-objective selection	Selection and export discipline
U.15	Legal postures for search	Legal/policy interpretation limits
U.16–U.19	Exact and heuristic implementations	Implementation and evaluation
U.20	Plan audit certificates	Artifact and verifier contract

Table 1: U-series spine after the RPLAN/RCTX fixed point.

3 Evidence Discipline

Claims are classified as fixture correctness, real-data smoke, benchmark result, empirical sweep, certification, or legal interpretation. A method may produce a valid plan or certificate without proving political fairness or legal compliance beyond its declared profile.

Evidence level	Supports	Does not support by itself
Fixture correctness	Deterministic behavior on toy cases	Real-data performance
Real-data smoke	CLI and data-path viability	Statistical generalization
Benchmark result	Measured runtime or quality on named runs	Universal dominance
Empirical sweep	Distributional comparison under a protocol	Legal compliance
Certificate	Declared constraints, lineage, or bounds	Political fairness
Legal interpretation	How evidence informs a posture	Technical validity alone

Table 2: Claim boundaries used by the U-series papers.

This discipline lets the implementation papers be useful before they are exhaustive. A staged exact solver can report fixture-level separation and audit lineage while marking broader runtime comparisons as future empirical work; a legal-posture paper can explain how a search record supports a claim without converting that record into a legal conclusion.

4 Current Implementation Boundary

The U-series should now be read through the package boundary established by U.10 and U.16–U.20. Search and optimization methods are not accepted merely because they produce plausible maps: final-plan claims need RPLAN/RCTX lineage, hash-bound manifests, verifier status, and a profile-scoped certificate.

Older papers in this track may describe algorithms as if they were complete CLI workflows. The current implementation boundary is narrower. Some methods are mature library substrates, some are audited vertical slices, and some remain methodological papers pending checked-in empirical packages. The claim boundary is therefore: search papers can explain candidate generation, diagnostics, and selection logic; they cannot by themselves prove legal sufficiency, statistical representativeness, or global optimality unless the cited package includes the corresponding verifier or solver certificate.

5 Limitations

This overview depends on the accuracy of the underlying paper drafts and implemented CLI surfaces. Future methods may require taxonomy revisions, and the paper should not imply that one evidence class can substitute for another.

The overview also inherits the limits of the fixed point it describes. RPLAN, RCTX, manifests, and audit certificates can make declared inputs, parameters, checks, and lineage inspectable. They do not certify that a plan is normatively fair, politically neutral, or legally sufficient outside the audit profile and recorded evidence.

6 Conclusion

U.0 provides the spine for the search and optimization papers: it makes roles, dependencies, and claim boundaries explicit before the more specialized U.12 through U.20 papers make their own contributions.

The next readiness step is cross-paper consistency: U.12 through U.15 should reuse this vocabulary for selection, certification, multi-objective export, and legal posture, while U.16 through U.20 should name the concrete crate, CLI, and artifact surfaces that instantiate it.

References

BISECT Project. U.0 search and optimization overview plan, 2026. BISECT internal paper plan.